

Data Science

Sem- III Project

Automated Survey Analysis

Individual report by-

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Team report by-

Pixel Plot Pro (Team- A)

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Team Report

Objective-

- To investigate the relationships between lifestyle choices, mental health, and well-being among individuals, with the aim of identifying patterns, factors influencing daily routines, and potential areas for intervention.
- The survey (Health and Wellness) aims to gather insights into sleep patterns, exercise habits, stress management strategies, and the impact of emotions on personal and social life.
- By analyzing this data, the goal is to inform strategies for promoting healthier lifestyles, better mental health, and overall well-being.



Approach-

1. Data Collection:

Collect diverse data related to health, mental status, and lifestyle choices from a broad population.

2. Data Preprocessing:

Ensure data cleanliness by handling missing values, converting data types, and addressing outliers.

3. Exploratory Data Analysis (EDA):

- ♣ Visualize and explore the distribution of variables.
- ♣ Identify patterns and correlations between health, mental status, and lifestyle choices.

4. Descriptive Statistics:

- ♣ Calculate descriptive statistics (mean, median, mode, etc.) for key variables.
- ♣ Analyze central tendencies and variabilities in the dataset.

5. Correlation Analysis:

- Conduct correlation analyses to quantify relationships between variables.
- Identify variables strongly associated with health and mental status.

6. Trend Identification:

- Identify trends or patterns over time.
- Explore how certain lifestyle variables may influence health and mental well-being trends.

7. Comparative Analysis:

Compare subgroups (e.g., different age groups, occupations) to identify differences in health and mental well-being based on lifestyle choices.

8. Insights and Recommendations:

- Derive insights based on descriptive and correlation analyses.
- Formulate recommendations based on observed trends and associations.
- Communicate insights about how lifestyle factors may predict or correlate with health and mental well-being across the population.



Target audience-

- This analysis is directed towards individuals of all age groups and various occupations who participated in the survey on lifestyle choices, mental health, and well-being.
- The insights gained from this diverse dataset aim to provide a comprehensive understanding of how different demographic groups perceive and manage their daily routines, stressors, and emotional well-being.

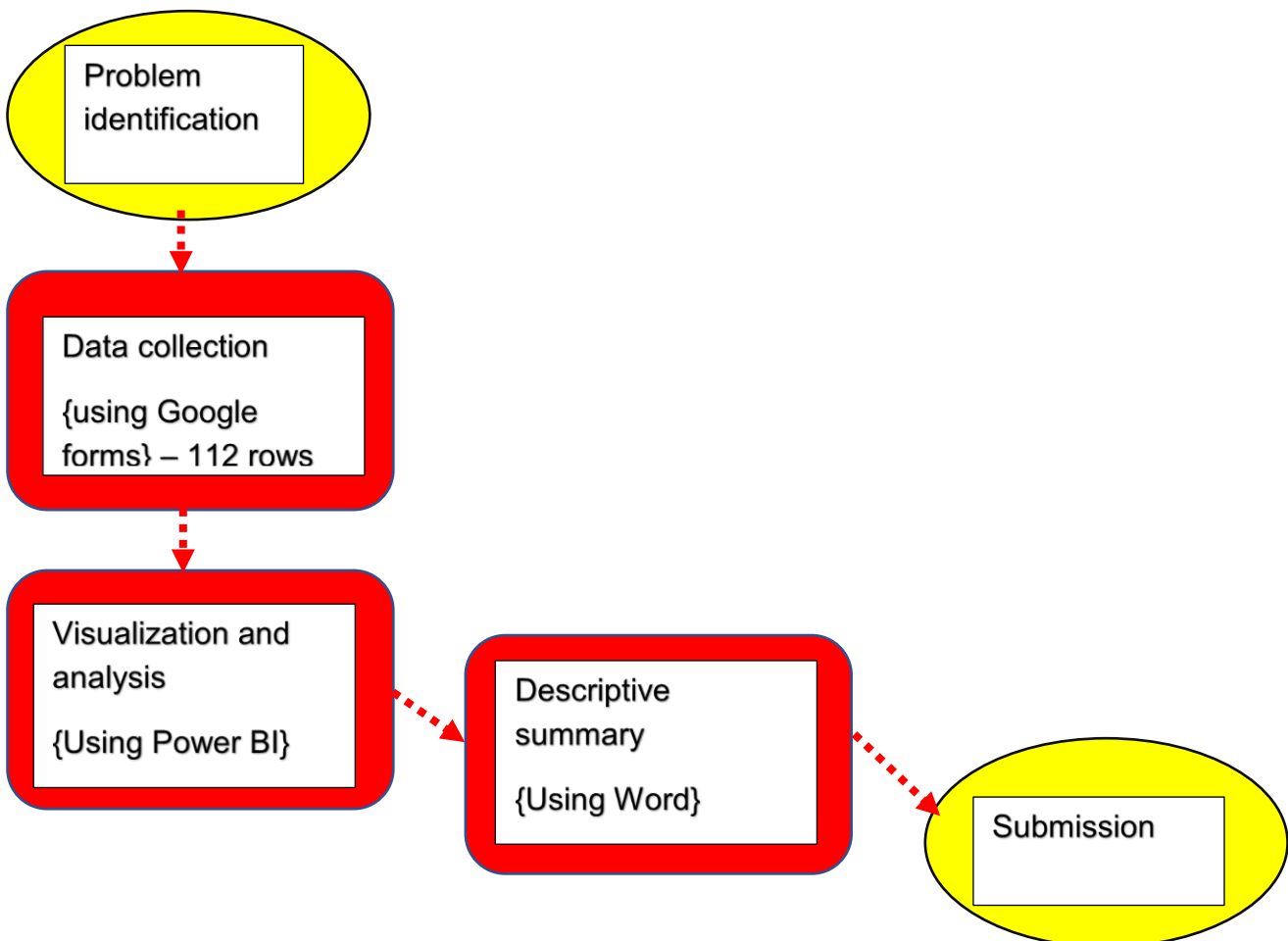


Type of data collected-

- 1) **Email (Text):** Identification and potential follow-up communication.
- 2) **Name (Text):** Identification.
- 3) **Date of Birth (Date):** Age calculation for age-related insights and tailored interventions.
- 4) **Gender (Nominal):** Demographic distribution insights for gender-specific analyses and targeted health strategies.
- 5) **Occupation (Nominal):** Understanding how different professions impact lifestyle and well-being, aiding in occupational health planning.
- 6) **Sleep Duration (Continuous):** Quantitative metric for understanding sleep patterns and their potential impact on overall health.
- 7) **Wake up Time (Time):** Precise temporal data for circadian rhythm analysis and insights into waking habits.
- 8) **Exercise Duration (Continuous):** Quantifies exercise habits, providing insights into physical activity levels.
- 9) **Screen Time (Continuous):** Quantitative measure of screen exposure, contributing to discussions on digital health.
- 10) **Fruits and Vegetables Servings (Discrete):** Quantifies dietary habits, contributing to nutritional assessments and recommendations.
- 11) **Source of Stress (Nominal):** Qualitative data on stressors, helping identify sources for targeted stress management strategies.
- 12) **Mental Health Status (Ordinal):** Graded assessment of mental well-being, aiding in the identification of trends and correlations.
- 13) **Importance of Religion and Spirituality (Ordinal):** Understanding the significance of religion and spirituality in individuals' lives for holistic well-being.

- 14) **Consistency of Sleeping (Nominal)**: Categorization of sleeping patterns for insights into the regularity of sleep routines.
- 15) **Emotion's Impact on Friends and Family (Nominal)**: Qualitative data on emotional impact, contributing to mental health and social well-being analyses.
- 16) **Daily Exercise Way (Nominal)**: Categorization of preferred exercise methods, aiding in personalized fitness recommendations.
- 17) **Deal with Stress (Nominal)**: Qualitative data on coping mechanisms, assisting in stress management strategies.
- 18) **Age (Continuous) {Calculated variable}**: Providing a continuous variable for age-related analyses and correlations.

Stages in our project-



Summary (Team)-

- ✂ The problem we took for our survey was related to Health and Wellness because, it is an interesting domain to learn, analyze, and visualize.
- ✂ It was interesting to conduct a survey by reaching out to everyone personally.
- ✂ The team expressed interest in exploring and understanding the dynamics of health and wellness, suggesting a genuine commitment to the survey's objectives.
- ✂ By focusing on the compelling aspects of health and wellness, the team aims to contribute meaningful insights to the field while enjoying the learning process.



Google survey form link-

[Health and Wellness Survey link](#)



Individual report

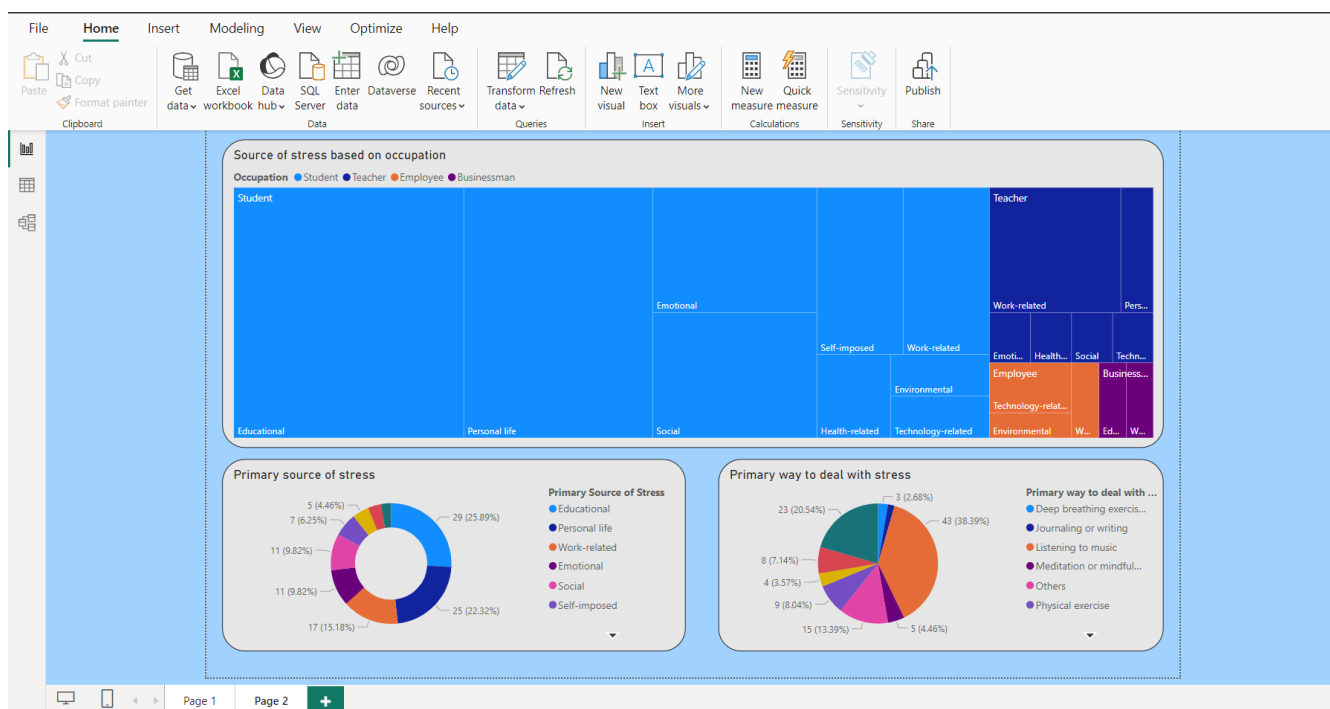
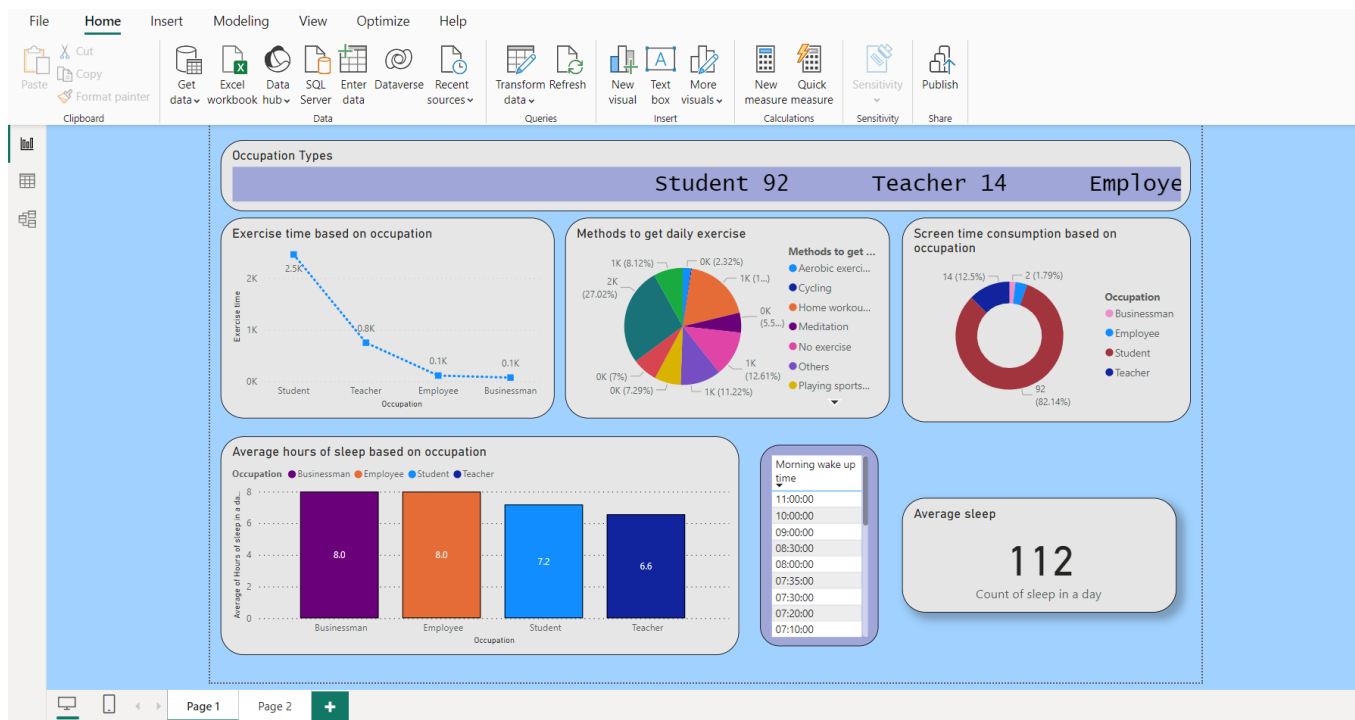
- The analysis is done with the help of a dashboard containing several visualizations that represent different case scenarios based on how the overall well-being of people is affected proportionally based on their occupations.
- The analysis was conducted using a Google form to collect the data where we found 4 different categories of people responding whose occupations were Student, Teacher, Employee, and Businessman, and the difference in their lifestyles defined their lifestyle.
- The analysis is divided into two halves of the dashboard. The first involves the analysis of general well-being like the amount of exercise they get in a day, what kind of exercise they do, their daily screen time intake, and the average amount of sleep consumption all based on their respective occupation. The second half mainly focuses on how the stress levels fluctuate based on the occupation. It consists of a graphical representation of the sources of stress within each occupation, what is the main primary source of stress amongst everyone, and the primary way to deal with stress. The dashboard also includes a slicer containing the count of the people of specific occupations where most are students.
- In the first half of the dashboard to represent the time of exercise done by each person of a respective occupation is represented using a line chart. The main reason for choosing this chart was to show a comparison for exercise done within all the occupations. Where we observe that the students tend

to get more exercise done as compared to the businesspersons who get very little to absolutely no exercise, which does affect their lifestyles.

- The next visualization was done on the popular methods in which people like to get their daily amount of exercise done. A pie chart was used as it was easy to compare amongst multiple types of workouts. It was observed that most people almost of about 27% prefer to either walk or jog to get their daily exercise and a very small number of people about 2% go for aerobic exercises and almost 12% of people prefer to not do any sort of exercise.
- Another visualization was a donut-based visualization that was used to compare the amount of screen time consumed by individuals of each occupation daily. It was observed that students mostly tend to spend most of their time on electronic devices and screen time consumption with about 92% of average screen time amongst all businessmen with very little screen time consumption of about 2%.
- The next visualization was about the average amount of sleep consumed by each individual based on their occupation. Here a bar chart was used to get a clear comparison between the four different types of occupations. It was observed that both businesspeople and regular employees get about 8 hours of sleep daily and student gets about 7 hours and a teacher gets about 6 hours of sleep making a teacher with the least amount of time consumed on sleeping.

- The second half of the dashboard consists of three visualizations based on the stress levels of personnel of each occupation. Where the first tree visualization depicted the different types of conditions that become a source of stress in each occupation and how each condition affects greatly. A donut chart was used to compare the different conditions of the primary source of stress where it was observed that educational stress was most common among all the responses about 29% of people mentioned that their primary source of stress was education, while 5% of people mentioned that their primary source of stress was work-related. The third visualization was a pie chart that gave us a comparison of how people tend to deal with their stressful situations. Where it was observed that from the several listed categories, about 43% of people say that listening to music is their primary way to deal with stress and about 3% of people say that deep breathing exercises are their way to deal with stress.
- In conclusion the entire dashboard gives a complete comparison on how the occupation of a person influences their overall health and mental well-being and what could be some of the popular ways to cope up with all the hinders caused.

Dashboard in Power BI Desktop



Dashboard in Power BI Mobile

